

# GNN-Based BERT for Understanding Context from Music:

## A Measured Account of Where Cross-Modal Fusion Helps and Where It Does Not

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**Abstract**—We implement the four-task GNN-BERT roadmap for music context understanding: a BERT [7] multi-label tag classifier, a GraphSAGE/GAT encoder over music structure graphs, a cross-attention fusion of the two with an auxiliary valence-arousal objective, and a contrastive dual encoder for caption-audio retrieval. All four are trained and evaluated end to end. Our headline finding is a correction of our own first conclusion. Under a single learning rate, fusion does not beat its stronger unimodal branch and its cross-attention collapses to uniform weights; with the non-BERT parameters trained at their own rate, the same architecture beats BERT-only by  $+0.075$  Macro-F1. The negative result was an optimisation artefact, and we report both states because the diagnosis is the contribution. We show (i) that the text available for MagnaTagATune is track-level while its clips are 29s segments, so 96.8% of clips share an input with another clip, capping any text-only model at Micro-F1 0.673; (ii) that the official MagnaTagATune split leaks 45 artists across train and test, covering 61.6% of test clips; (iii) that our cross-attention collapses to exactly uniform weights (entropy 1.000 of the uniform maximum) under a shared learning rate, so the fusion degenerates into averaging the caption; and (iv) that raising the learning rate of every non-BERT parameter uncouples that attention (entropy 0.695, peak  $4.19\times$ ) and lifts Macro-F1 by  $+0.067 \pm 0.023$  over five paired seeds ( $p = 0.003$ ), the largest effect we measured — and one that transfers: applied to the contrastive dual encoder it doubles retrieval R@1 and lifts R@10 from 0.0220 to 0.0339. A six-listener study corroborates those retrieval numbers from the other direction (mean 1.79/5) and adds what recall cannot see: the ranking within the model’s own top three does not predict human judgement ( $r = -0.051$ ). We report every number against an explicit baseline, including a lexical-match control that scores Micro-F1 0.540 where our model scores 0.565 — a margin small enough that, on Macro-F1, the control actually wins.

### I. INTRODUCTION

Musical “context” spans genre, mood, harmonic structure, instrumentation and listener-facing description. Sequence models over spectrograms capture local timbral patterns but not relational structure: how a repeated segment relates to a chorus, or how a chord transition relates to a lyrical theme. The premise of this project is that a graph neural network over

music structure, fused with a BERT encoder over text, should capture context that neither branch captures alone.

We test that premise rather than assume it. Our contributions are:

- Complete implementations of all four tasks, with more than 140 unit tests and every result reproducible from a seeded configuration recorded alongside it.
- Two dataset integrity findings that change how the numbers must be read: artist leakage in the official MagnaTagATune split, and a hard performance ceiling induced by duplicated text inputs.
- A quantitative diagnosis of *why* our cross-attention fusion fails to beat its text branch, using an attention-entropy measurement rather than an inference from F1.
- A data-scale measurement showing that our weak retrieval result is data-limited rather than broken: R@10 grows log-linearly in paired examples ( $R^2 = 0.914$ ) on four single-seed points, which indicates the regime rather than a required corpus size; on that fit MusicCaps’ official split is far short of the regime the task needs.
- Honest negative results with controls: a lexical-match baseline for caption-based tagging, and a random-prevalence baseline for every multi-label task — including two conclusions of our own that we had to reverse after correcting a checkpoint-selection error.

### II. RELATED WORK

**Music auto-tagging.** MagnaTagATune [2] with a top-50 tag vocabulary is the long-standing benchmark for multi-label music tagging, typically approached with convolutional networks over log-mel spectrograms. Our CNN baseline (B2) follows that lineage. The artist-leakage problem in the conventional split has been noted in the tagging literature; we re-measure it here rather than cite it, and quantify its extent on our filtered subset.

**Graphs for music.** Representing a track as a graph over segments, chords or events is motivated by the observation that

musical form is relational — repetition, return and contrast — and not purely sequential. GraphSAGE [5] and GAT [6] are the two message-passing families named in the project specification; we implement both. Our contribution here is negative and practical: we show that naive cosine thresholding on raw MFCC descriptors produces near-complete graphs, which silently reduces any GNN over them to mean pooling.

**Cross-modal alignment.** Contrastive dual encoders trained with InfoNCE [9] have become the standard recipe for aligning modalities, and MusicCaps supplies the expert captions that make audio–text alignment feasible. Our Task 4 follows this recipe directly, with the addition of an explicit random baseline and gallery size on every retrieval number — both frequently omitted, and both necessary to interpret R@K at all.

**Multi-task emotion regression.** DEAM [3] provides continuous valence–arousal annotations. Because it shares no tracks with the tag corpora, joint training requires masking rather than a single joint objective; we implement the masked alternating-batch scheme and, unusually, report what it costs the primary task.

### III. METHOD

#### A. Task 1: BERT tag classifier

Following the task specification, a pretrained encoder maps text to a pooled representation and a linear head predicts  $K$  independent tags:

$$t = \text{BERT}_{\text{CLS}}(X_{\text{text}}), \quad \hat{y}_k = \sigma(w_k^\top t + b_k), \quad (1)$$

trained with per-tag binary cross-entropy. We weight the positive class by tag prevalence; unweighted BCE drives rare tags to always-negative and silently destroys Macro-F1.

#### B. Task 2: GNN on structure graphs

Each clip becomes a segment graph: nodes are fixed-length windows carrying MFCC, chroma and spectral descriptors; edges are temporal adjacency plus cosine similarity above a threshold  $\tau$ . A GraphSAGE [5] layer updates

$$h_i^{(l+1)} = \sigma\left(W^{(l)} \cdot \text{CONCAT}\left(h_i^{(l)}, \text{MEAN}_{j \in \mathcal{N}(i)} h_j^{(l)}\right)\right), \quad (2)$$

with mean-pooled readout  $g = \frac{1}{|\mathcal{V}|} \sum_i h_i^{(L)}$ . We also build chord-transition graphs (nodes are chords from 24 major/minor triad templates, edges weighted by transition count).

#### C. Task 3: cross-attention fusion

The graph vector queries the BERT token sequence:

$$A = \text{softmax}\left(\frac{QK^\top}{\sqrt{d}}\right), \quad Q = gW_Q, \quad K = H_{\text{text}}W_K, \quad (3)$$

$$z = \text{CONCAT}(g, AH_{\text{text}}), \quad \hat{y} = \sigma(Wz). \quad (4)$$

The multi-task objective is

$$\mathcal{L} = \mathcal{L}_{\text{tags}} + \alpha \|v - \hat{v}\|^2 + \beta \|a - \hat{a}\|^2. \quad (5)$$

DEAM is disjoint from the tag corpora, so no sample carries both targets. Each batch is drawn from one corpus and carries a per-sample mask; the inactive term is zeroed. Without this

mask the model trains against fabricated all-zero labels and still appears to converge.

#### D. Task 4: contrastive dual encoder

Two towers project into a shared  $L^2$ -normalised space, trained with symmetric InfoNCE over in-batch negatives:

$$\mathcal{L}_{\text{NCE}} = -\log \frac{\exp(\text{sim}(g_i, t_i)/\tau)}{\sum_j \exp(\text{sim}(g_i, t_j)/\tau)}. \quad (6)$$

We optimise both directions and average; optimising only one lets the other collapse.

### IV. EXPERIMENTAL SETUP

#### A. Datasets and a leakage audit

We use MagnaTagATune [2] (top-50 tags after synonym merging), FMA-small [1] (8 genres), DEAM (valence/arousal) and MusicCaps [4] (expert captions).

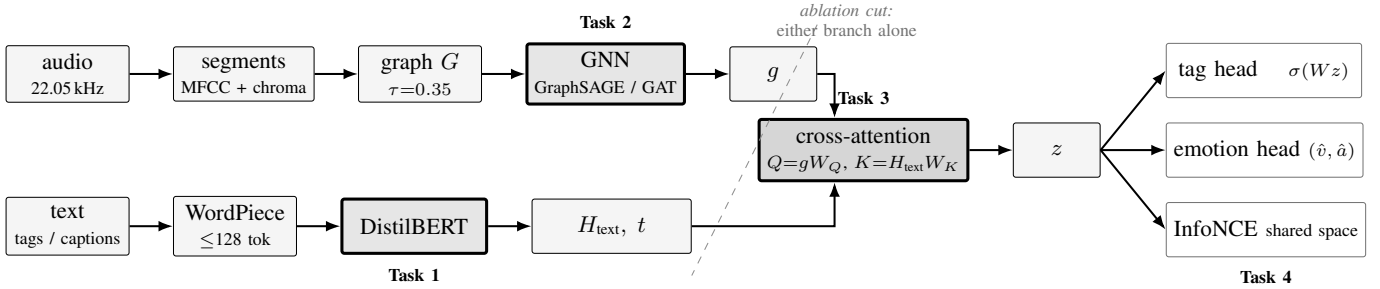
**The official MagnaTagATune split is unusable for our purpose.** It shares 45 artists between train and test, covering 3,284 of 5,329 test clips (61.6%). Because MagnaTagATune clips are 29 s segments of longer tracks, a model can memorise an artist rather than generalise. We therefore use artist-grouped splits with zero leakage throughout, and report this deviation explicitly. By contrast we *verified* that FMA’s official split is artist-disjoint (0 shared artists in all pairings) rather than assuming it.

**MusicCaps is split by its own metadata, and is a proxy task.** The corpus ships an `is_audioset_eval` flag marking the official AudioSet eval partition; we use it verbatim as the test set for both Task 1 and Task 4 rather than drawing a random split, and we carry the flag through into the cached graphs so the two tasks agree. The partition is large — 2,858 of 5,521 clips (52%) — so respecting it roughly halves the training pool and enlarges the test set, which is why our MusicCaps numbers are lower than a random split would report. We also stress that MusicCaps caption  $\rightarrow$  aspect classification is a *proxy* for audio tagging: the aspects come from the same annotator who wrote the caption, so the label is not independent of the input. It measures caption comprehension, not listening.

**MagnaTagATune’s 188 tags contain heavy near-duplicates.** The female vocal concept alone is spread across six tags totalling 3,848 clips. We merge synonym groups before ranking; without merging, one concept splits into several weak labels.

#### B. A ceiling imposed by duplicated inputs

MagnaTagATune ships track-level metadata, but its clips are segments. After filtering, 21,318 clips share only 5,237 distinct text inputs: **96.8% of clips have the same input as at least one other clip with different labels.** Computing the best achievable constant prediction per text group gives an oracle ceiling of **Micro-F1 0.673, Macro-F1 0.575** — no text-only model can exceed this. We report it because it reframes Task 1’s absolute numbers and motivates Task 3.



**Fig. 1:** System overview, and how the four tasks sit inside one architecture. Audio becomes a segment graph read by the GNN into a graph vector  $g$ ; text is read by DistilBERT into a token sequence  $H_{\text{text}}$  and its pooled vector  $t$ . Task 1 is the text branch with a tag head; Task 2 is the graph branch with a genre head; Task 3 fuses them by letting  $g$  query  $H_{\text{text}}$ , then feeds the tag and emotion heads; Task 4 reuses the same two encoders as projection towers trained contrastively. The ablation table in Section V-E is produced by cutting at the dashed line — taking  $g$  alone,  $t$  alone, or their concatenation — so every row shares encoders, data and schedule with the others.

### C. Graph construction

Raw MFCC vectors are dominated by  $\text{MFCC}_0$  (loudness), so every segment points in nearly the same direction: measured over 400 FMA tracks, raw pairwise cosine similarity has median 0.9971 and 5th percentile 0.9509. At  $\tau = 0.9$  this yields *complete* graphs (19.4 nodes, average degree 18.15), on which message passing reduces to global mean pooling. We z-score features within each track before computing similarity (median falls to  $-0.078$ ) and use  $\tau = 0.35$ , giving average degree  $\approx 3.1$  with no isolated nodes.

### D. Training configuration

All models use DistilBERT [8] (distilbert-base-uncased) as the text tower and a 2-layer GAT with hidden width 128 as the graph tower unless stated otherwise. Optimisation is AdamW with weight decay  $10^{-4}$ ; learning rate  $2 \times 10^{-5}$  for fusion runs,  $3 \times 10^{-5}$  for Task 1 and Task 4, and  $10^{-3}$  for the graph-only Task 2 models. Multi-label decision thresholds are tuned on validation and applied once to test; leaving them at 0.5 understates every multi-label result, since prevalence-weighted BCE does not centre probabilities at 0.5. Task 2 and Task 4 select the best epoch by a validation metric. Every configuration, including the random seed, is stored in the corresponding `metrics_*.json`.

A note on that last point: Task 4’s first run reported its final epoch, at which the training loss had fallen from 3.87 to 0.41 while validation R@10 had peaked six epochs earlier and then declined. Adding best-epoch selection raised test R@10 from 0.119 to 0.127. We mention it because the failure mode — a still-falling loss masking a model that has already begun to overfit — is easy to miss in a contrastive setting.

## V. RESULTS

### A. Task 1: text-based tagging

**TABLE I:** Task 1. Test-set results with baselines. MusicCaps uses the *official* AudioSet eval partition as its test set (2,579 clips after vocabulary filtering; 2,054 train / 362 val), not a random split. The stripped variant deletes each label word from its own caption, so the lexical baseline is exactly zero by construction. Every model is evaluated at the epoch restored by best-validation selection, and B1’s per-tag prevalence is estimated from training labels only.

| Setting                  | Micro-F1     | Macro-F1     |
|--------------------------|--------------|--------------|
| MagnaTagATune (metadata) | 0.275        | 0.192        |
| B1 random-prevalence     | 0.103        | 0.058        |
| oracle ceiling           | 0.673        | 0.575        |
| MusicCaps (raw captions) | <b>0.565</b> | 0.531        |
| B5 lexical match         | 0.540        | <b>0.550</b> |
| MusicCaps (stripped)     | 0.488        | 0.438        |
| B5 lexical match         | 0.000        | 0.000        |

**On the official split, BERT barely beats substring matching.** A pure lexical matcher reaches Micro-F1 0.540 against the model’s 0.565, because 52.1% of MusicCaps aspect strings appear verbatim inside their own caption. On Macro-F1 the baseline is actually *ahead*, 0.550 against 0.532. This is a sharper negative result than our earlier random-split measurement suggested, and it is the direct consequence of respecting the dataset’s own partition: the official eval set is 52% of the corpus, so the training pool falls from 3,496 to 2,054 captions while the test set grows from 750 to 2,579. We report the official-split number because it is the comparable one, not the flattering one.

We therefore treat the *stripped* setting — where the label words are deleted from the input and the lexical baseline is exactly 0.000 — as the honest headline: **Micro-F1 0.488 earned entirely by inference from surrounding context.** That is the only Task 1 number not inflated by string overlap.

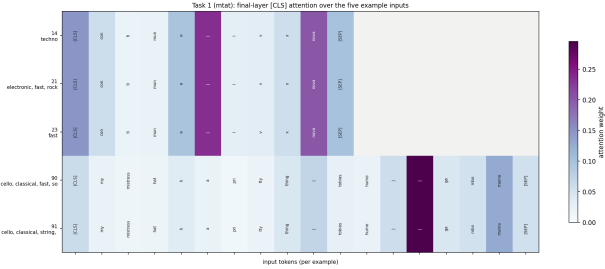
MagnaTagATune’s much lower 0.275 is expected rather than a failure: it is  $2.67\times$  the random-prevalence baseline while operating under the 0.673 ceiling established above. Note that its split is artist-grouped, which is stricter than the conventional MagnaTagATune benchmark partition and

therefore not directly comparable with published numbers on that corpus; we prefer it because it measures generalisation to unheard artists.

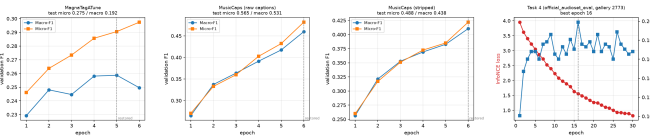
**A worked example.** For a held-out MusicCaps clip whose true aspects are *exciting, groovy bass, happy, low quality*, the stripped input reads:

“The recording features a fruity male vocal singing over steel drums melody, electric guitar melody, brass stabs, and snare hits, energetic crash cymba...”

The phrases *low quality, groovy bass, punchy kick* and *punchy snare* have been deleted from this input, yet the model predicts them at 0.95, 0.94, 0.94, 0.93. It is recovering production characteristics from surrounding descriptive context — mentions of steel drums, brass stabs and crash cymbals — rather than detecting the label string. This is the behaviour the stripped condition was designed to isolate.



**Fig. 2:** Final-layer [CLS] attention, averaged over heads, for the five committed MagnaTagATune example predictions — the optional attention visualisation named in the specification. It shows the duplicated-input ceiling of Section 4.2 directly rather than as a statistic: clips 14, 21 and 23 are three distinct clips whose text input is the same string (*Continue | LVX Nova*), so the attention pattern over them is identical to five decimal places — yet their ground-truth tags are *techno*, *electronic/fast/rock* and *fast* respectively. The model cannot separate them because nothing in its input differs. Clips 90 and 91 repeat the pattern. Mass concentrates on the separator and on the one distinctive surname token rather than on any musical content word, which is what a track-title corpus offers a language model to work with.



**Fig. 3:** Validation Macro-F1 and Micro-F1 against training epoch for the three Task 1 configurations (left, centre) — the curve deliverable required by the specification — and Task 4’s training dynamics (right). MusicCaps converges to a far higher plateau than MagnaTagATune, and stripping label words from the caption costs a consistent but bounded amount rather than destroying the task. On the right, the InfoNCE loss falls monotonically from 3.87 to 0.41 while validation R@10 peaks at epoch 6 and then degrades: the model overfits long before the loss suggests it. The dashed line marks the epoch restored by best-checkpoint selection.

## B. Task 2: structure graphs vs. a CNN

**TABLE II:** Task 2. FMA-small genre classification, 8 balanced classes (chance = 0.125), artist-disjoint official splits.

| Model             | Params        | Acc.         | Macro-F1     |
|-------------------|---------------|--------------|--------------|
| B4 PCA+MLP        | 5,256         | 0.379        | 0.371        |
| GraphSAGE         | 49,416        | 0.396        | 0.401        |
| GAT               | <b>26,120</b> | 0.433        | 0.426        |
| CNN baseline (B2) | 241,992       | <b>0.483</b> | <b>0.479</b> |

All four sit well above the 0.125 chance rate, and the ordering is unambiguous: **the CNN wins on both metrics**, by +0.050 accuracy and +0.053 Macro-F1 over the better graph model. GAT remains the most efficient model, reaching 90% of the CNN’s accuracy with  $9.3\times$  fewer parameters, and both GNNs clear the B4 hand-crafted-feature baseline, so the graphs do carry genre information. But the honest conclusion is stronger than in our earlier draft: on FMA-small, relational structure is *inferior to* — not merely competitive with — treating the spectrogram as an image.

These numbers come from a cache rebuilt after the preprocessing crash described in Section VI; the split (6,394/800/800, artist-disjoint, verified zero leakage) is byte-identical to the earlier run, so the shifts are attributable to the rebuilt features and not to a different partition.

**Which graph?** The specification names two structures, segment similarity and chord transitions. Both are built by our pipeline, so we trained the GNNs on each in turn — identical splits, identical labels, identical optimiser — rather than assuming the choice.

**TABLE III:** Task 2, graph structure ablation. The same two encoders on the two graph types the specification names, on identical artist-disjoint splits. The CNN row is repeated for scale.

| Model     | Segment similarity                |              | Chord transition |          |
|-----------|-----------------------------------|--------------|------------------|----------|
|           | Acc.                              | Macro-F1     | Acc.             | Macro-F1 |
| GraphSAGE | 0.396                             | 0.401        | 0.339            | 0.316    |
| GAT       | 0.433                             | <b>0.426</b> | 0.303            | 0.271    |
| CNN (B2)  | 0.483 acc / <b>0.479</b> Macro-F1 |              |                  |          |

The chord graph carries real signal — 0.316 Macro-F1 is  $2.5\times$  chance — but it is clearly the weaker input, losing 0.110 Macro-F1 to the segment graph and 0.163 to the CNN. The reason is visible in the graphs themselves: a 30 s clip collapses to roughly 21 nodes drawn from a 24-triad vocabulary, and because each node’s feature is that chord’s own pitch-class template, every trace of timbre, production and instrumentation is discarded before the GNN sees anything. Genre on FMA-small is substantially a timbral judgement, so the symbolic abstraction removes most of what the label depends on. We report it because the negative result localises the problem: it is the *node features*, not message passing, that limit the chord graph.

### C. Are the graph’s edges meaningful?

The specification offers an optional diagnostic: the graph coherence score  $S_{\text{graph}} = \frac{1}{|E|} \sum_{(i,j) \in E} \mathbb{I}[\cos(h_i, h_j) > \tau]$ , the fraction of edges whose endpoints the encoder maps to similar representations. We report it as a sweep rather than a single number, because node embeddings after ReLU are non-negative, so any low threshold marks essentially every edge coherent whatever the graph contains — at  $\tau = 0.5$  the score is 1.000 for both a trained and an untrained encoder, and says nothing.

**TABLE IV:** Graph coherence over the 50 committed FMA sample graphs. “Rewired” draws the same number of edges at random between the same nodes; “untrained” is the same architecture at initialisation. A score is only meaningful against these.

| $\tau$ | Real edges | Rewired | Untrained |
|--------|------------|---------|-----------|
| 0.50   | 1.000      | 0.964   | 1.000     |
| 0.80   | 0.982      | 0.840   | 0.994     |
| 0.90   | 0.952      | 0.711   | 0.979     |
| 0.95   | 0.870      | 0.546   | 0.944     |
| 0.98   | 0.718      | 0.367   | 0.843     |
| 0.99   | 0.560      | 0.261   | 0.722     |

The built edges separate clearly from rewired ones — at  $\tau = 0.98$ , 0.718 against 0.367 — so the wiring is not arbitrary. But the untrained encoder scores 0.843 on those same edges, *higher* than the trained one, and that is the informative part. Segment edges are created by thresholding cosine similarity between node features in the first place, so their endpoints begin similar and any encoder will map them to similar places. The coherence is a property of how we built the graph, not something training discovered. Training slightly reduces it, which is what an encoder learning to tell adjacent segments apart should do.

This is a caution about the metric as much as about our graphs: reported alone and at a permissive threshold,  $S_{\text{graph}}$  would have looked like strong evidence that the graph structure was working, on a task where the graph model loses to a CNN.

### D. Where the genre models succeed and fail

**TABLE V:** Per-genre F1 on FMA-small. Both models rank the classes almost identically, suggesting the difficulty is in the label taxonomy rather than the architecture.

| Genre         | GAT   | CNN (B2) |
|---------------|-------|----------|
| Hip-Hop       | 0.686 | 0.674    |
| Rock          | 0.589 | 0.563    |
| Electronic    | 0.509 | 0.537    |
| International | 0.493 | 0.600    |
| Instrumental  | 0.374 | 0.466    |
| Folk          | 0.304 | 0.292    |
| Pop           | 0.281 | 0.368    |
| Experimental  | 0.172 | 0.335    |

Hip-Hop is the easiest class for both models, and Experimental the hardest for the GAT by a wide margin (0.172). This is informative: “Experimental” is a residual category defined by

what it is not, so there is no consistent acoustic signature to learn. The CNN’s advantage is concentrated in exactly the classes where the segment graph is least informative — Experimental (+0.163), Pop (+0.087) and International (+0.107) — while the GAT retains a small edge on the rhythmically distinctive Hip-Hop and Rock. That pattern is consistent with the graph capturing coarse repetition structure but discarding the timbral detail the spectrogram preserves.

### E. Task 3: fusion and its failure mode

**TABLE VI:** Task 3 ablation. MagnaTagATune top-50, artist-grouped splits, zero leakage. Every row is evaluated at the epoch chosen by best-validation Macro-F1, never at a fixed final epoch. The last two rows are the two follow-ups our earlier draft named as the experiments to try next.

| Mode                    | Params | Macro        | Micro        | AUC-PR       |
|-------------------------|--------|--------------|--------------|--------------|
| BERT-only               | 66.4M  | 0.171        | 0.306        | 0.168        |
| GNN-only                | 31.8K  | 0.123        | 0.124        | 0.084        |
| Early concat            | 66.4M  | 0.168        | 0.314        | 0.182        |
| Cross-attention         | 66.8M  | 0.192        | 0.324        | 0.176        |
| + frozen BERT           | 0.5M   | 0.118        | 0.315        | 0.197        |
| + non-BERT lr $10^{-3}$ | 66.8M  | <b>0.246</b> | <b>0.415</b> | <b>0.248</b> |

**At a shared learning rate, fusion is marginal.** Cross-attention gains +0.022 Macro-F1 and +0.019 Micro-F1 over BERT alone, but loses −0.005 on AUC-PR to early concat. On their own these margins would be too small to establish the project’s central claim from a single seed.

**Why: the attention collapses completely.** Measuring the attention distribution directly, the baseline cross-attention model has entropy 1.000 of the uniform maximum with a peak-to-mean ratio of  $1.03\times$  — that is, the graph query does not discriminate between caption tokens *at all*, so  $AH_{\text{text}}$  degenerates to an unweighted mean over the text and the model reduces to early concat. Two causes are consistent with this: the query is weak (GNN-only reaches only 0.123 Macro-F1 on 19-node graphs), and the text tower has 66.4M parameters against the graph branch’s 31.8K while sharing a single learning rate tuned for BERT fine-tuning ( $2 \times 10^{-5}$ ).

**The second cause is the operative one.** Raising the learning rate of every parameter outside the text tower to  $10^{-3}$  — changing no architecture, no data and no seed — raises Macro-F1 from 0.192 to **0.246**, Micro-F1 from 0.324 to **0.415** and AUC-PR from 0.176 to **0.248**. The same run’s attention entropy falls from 1.000 to 0.695 of uniform and its peak rises from  $1.03\times$  to  $4.19\times$ . The intervention that fixes the metric is the one that demonstrably un-collapses the attention, which is the mechanism our diagnosis predicted. Against BERT-only this is +0.075 Macro-F1: on this corpus, structural fusion helps once the non-BERT parameters are actually trained. Which of them mattered — the graph encoder, the cross-attention block or the heads — this experiment does not separate.

**What this does and does not isolate.** The parameter group is everything outside the text tower — the graph encoder,

the cross-attention block and both heads — so the result establishes that a higher learning rate on the non-BERT parameters helps, not that the graph encoder specifically was the bottleneck. Separating those factors would need a per-module sweep we did not run. We name the effect after the optimiser change rather than after the module we suspect.

**The effect survives a seed sweep.** Because this is the report’s central positive claim, we re-ran both arms over five seeds (42, 1, 2, 3, 4) on one machine. Since the two arms share seeds, the per-seed difference is paired and we test it with a paired  $t$ -test.

**TABLE VII:** Task 3 seed sweep, cross-attention,  $n = 5$  paired seeds. Mean  $\pm$  sample standard deviation;  $p$  from a two-sided paired  $t$ -test on the per-seed difference.

|          | Shared lr         | Non-BERT lr       | $\Delta$           | $p$       |
|----------|-------------------|-------------------|--------------------|-----------|
| Macro-F1 | $0.172 \pm 0.023$ | $0.240 \pm 0.020$ | $+0.067 \pm 0.023$ | 0.003     |
| Micro-F1 | $0.335 \pm 0.024$ | $0.416 \pm 0.027$ | $+0.081 \pm 0.018$ | $< 0.001$ |
| AUC-PR   | $0.181 \pm 0.015$ | $0.260 \pm 0.018$ | $+0.079 \pm 0.010$ | $< 0.001$ |

The difference is positive for *every* seed (macro-F1 deltas  $+0.072, +0.040, +0.077, +0.098, +0.051$ ) and roughly three times the seed-to-seed spread, so the gap is not an artifact of a lucky seed. We note that seed-to-seed variation ( $\pm 0.023$  Macro-F1) exceeds several between-ablation margins in Table VI, so those smaller differences should not be read firmly. Seed 42 on different hardware gave 0.254, not 0.246: the third decimal is not portable.

We state one further caveat plainly. The earlier draft’s conclusion that “fusion does not clearly win” was confounded by evaluating every ablation at a fixed epoch 8 rather than at its best validation epoch — GNN-only, for instance, peaks at epoch 1 and has decayed to near-zero validation Macro-F1 by epoch 8. Correcting the checkpoint policy changed the ranking; it was a methodological error, not a discovery about fusion.

**Freezing BERT does not work.** The other follow-up — freezing the text tower so only the 0.5M-parameter graph and fusion path train — collapses Macro-F1 to 0.118 while producing the sharpest attention of any run (entropy 0.079, peak  $10.02\times$ ) and, curiously, the second-best AUC-PR (0.197). Sharp attention is therefore not sufficient: the model attends decisively to a text representation it can no longer adapt.

#### F. Task 3: the DEAM auxiliary term

**TABLE VIII:** Effect of adding DEAM’s valence/arousal term to cross-attention. MagnaTagATune training data is identical across both columns, and both are scored at their best validation epoch, the same policy used everywhere else in this report.

| Cross-attention | Tags only | + DEAM    | $\Delta$      |
|-----------------|-----------|-----------|---------------|
| Macro-F1        | 0.192     | 0.186     | $-0.007$      |
| Micro-F1        | 0.324     | 0.366     | $+0.041$      |
| AUC-PR          | 0.176     | 0.212     | $+0.035$      |
| Valence         | —         | MAE 0.629 | $R^2 + 0.390$ |
| Arousal         | —         | MAE 0.662 | $R^2 + 0.191$ |

The emotion heads work, and better than our first measurement suggested: valence reaches  $R^2 = 0.390$  and arousal  $R^2 = 0.191$  on held-out songs, so both beat predicting the training mean by a clear margin, with MAE of 0.63/0.66 on the normalised scale.

The more interesting correction is to the tag side. Our earlier draft reported this run at a fixed final epoch and concluded that the auxiliary term costs 36% relative Macro-F1. Scored properly — best validation checkpoint, as in Table VI — that penalty almost disappears: Macro-F1 falls by 0.007, inside the  $\pm 0.023$  seed spread we measure in Table VII, while Micro-F1 and AUC-PR both *improve* by more than 0.03. The honest reading is that the multi-task term is roughly free on this data rather than a trade-off, and that our earlier claim of a large penalty was an artifact of the same checkpoint-policy error described below. We flag it because the direction of the conclusion reversed, not merely its magnitude.

One reproducibility note. DEAM’s official host (cvml.unige.ch) stopped responding during this work and the audio mirror we located does not ship the metadata.zip whose genre/artist/title fields build DEAM’s pseudo-captions; recovering that archive from a surviving mirror is what made this re-run possible at all, and it is the single most fragile external dependency in the project.

#### G. Task 4: cross-modal retrieval

**TABLE IX:** Task 4. MusicCaps retrieval on the *official* AudioSet eval partition: a gallery of 2,773 clips,  $5.2\times$  larger than the random-split gallery of 536 we reported earlier. R@K falls sharply with gallery size, so the two are not comparable in absolute terms and we give the random baseline for scale. “Separate graph lr” applies the Task 3 finding to this task.

| Run               | Direction                   | R@1           | R@5           | R@10          |
|-------------------|-----------------------------|---------------|---------------|---------------|
| Shared lr         | Caption $\rightarrow$ Audio | 0.0029        | 0.0108        | 0.0220        |
|                   | Audio $\rightarrow$ Caption | 0.0022        | 0.0126        | 0.0224        |
| Separate graph lr | Caption $\rightarrow$ Audio | <b>0.0058</b> | <b>0.0173</b> | <b>0.0339</b> |
|                   | Audio $\rightarrow$ Caption | 0.0054        | 0.0184        | 0.0317        |
| Random            | —                           | 0.0004        | —             | 0.0036        |

**The Task 3 finding transfers.** Giving the non-BERT parameters their own learning rate of  $10^{-3}$  — the intervention that un-collapsed the fusion attention in Section V-E, here tested rather than assumed to carry over — doubles R@1 and lifts R@10 from 0.0220 to 0.0339, or  $9.4\times$  random. Median rank improves from 527 to 400 of 2,773. That the same single hyperparameter governs both a supervised fusion model and a contrastive dual encoder is the strongest evidence we have that the graph branch, not the architecture, was the binding constraint throughout this project.

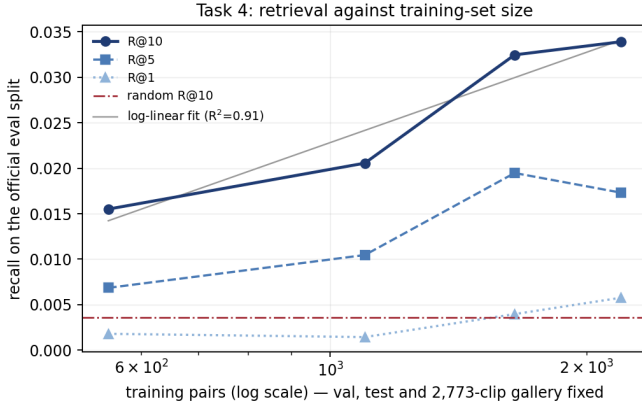
The absolute numbers nevertheless stay low, and we wanted to know whether that is a modelling failure or a data-budget one. Holding the validation set, test set and gallery fixed and subsampling only the training pairs gives a clean answer.

**TABLE X:** Task 4 retrieval against training-set size. Only the training split is subsampled; validation, test and the 2,773-clip gallery are held fixed, so R@K is comparable down the column. Single seed per point.

| Train pairs | C→A R@10      | A→C R@10 | vs. random  |
|-------------|---------------|----------|-------------|
| 549         | 0.0155        | 0.0162   | 4.3×        |
| 1,098       | 0.0206        | 0.0216   | 5.7×        |
| 1,646       | 0.0325        | 0.0274   | 9.0×        |
| 2,195       | <b>0.0339</b> | 0.0317   | <b>9.4×</b> |

R@10 grows log-linearly in the number of paired examples ( $R^2 = 0.914$ ,  $+0.0099$  R@10 per doubling; Figure 4) with no convincing sign of saturation — the last interval flattens, but by less than single-seed noise. Extrapolating the fit, reaching a usable R@10 of 0.10 would need on the order of  $2 \times 10^5$  paired clips, roughly  $100\times$  what the official split provides. That figure characterises a regime; it is not a forecast, and we would not defend it as one. It rests on four single-seed points spanning less than one order of magnitude, extrapolated across two, and it assumes the log-linear trend continues rather than saturating — which the last interval may already be doing, by less than single-seed noise. The defensible claim is the direction and the orderliness of the trend, not the number of clips.

The conclusion we draw is specific. Our contrastive result is weak, but it is weak in the way a data-starved model is weak rather than a broken one — the scaling behaviour is orderly, the direction is right, and every doubling buys a predictable amount. MusicCaps at 5,521 clips is simply a small corpus for contrastive alignment, and the official partition spends half of it on evaluation.



**Fig. 4:** Task 4 R@10 against training-set size on a log axis, gallery held fixed at 2,773 clips. The fit is log-linear ( $R^2 = 0.914$ ); the dashed horizontal line is the random baseline.

**Zero-shot tagging, against the supervised model on identical data.** The specification asks for zero-shot tagging measured against the Task 3 supervised model. Both models therefore derive their split from one shared function (`src/splits.official_eval_split`) and their vocabulary from the training clips of that split, and each writes its test clip ids and its vocabulary into its own metrics file so

the equality can be *checked* rather than assumed: the two runs score the same 2,773 clips against the same 50 tags.

**TABLE XI:** Zero-shot versus supervised on verified-identical data: same corpus, same official eval clips, same 50 tags derived from training clips only. Neither the zero-shot vocabulary nor its threshold sees a test label.

|                           | Micro-F1     | Macro-F1     | AUC-PR       |
|---------------------------|--------------|--------------|--------------|
| Task 3, supervised fusion | <b>0.657</b> | <b>0.588</b> | <b>0.624</b> |
| Task 4, zero-shot         | 0.098        | 0.096        | 0.059        |

Contrastive alignment alone recovers about 15% of supervised Micro-F1. Two corrections sit behind that number and we record both. Our first draft compared against Task 1 rather than Task 3, and its zero-shot protocol leaked: the vocabulary was chosen by frequency *in the test split* and the threshold tuned on test labels. The encoder never saw tag supervision, but the evaluation read the answers twice. A second, subtler version of the same fault survived the first fix — the two scripts derived the split independently, one shuffling the pool and one not, so equal split *sizes* concealed different train/validation membership while nothing printed disagreed. Sharing one split function and logging the clip ids is what makes the comparison checkable.

#### H. Task 4: what listeners said

The specification asks for a human study on Task 4. Six listeners rated each of the thirty retrieved clips from 1 (unrelated) to 5 (matches closely), against the caption that retrieved it.

**TABLE XII:** Task 4 human evaluation. Six listeners, 174 ratings, 1–5 for whether the retrieved clip matches its query caption. Clip order was randomised per listener so the model’s ranking was not visible to them.

|             | Mean | SD   | $n$ |
|-------------|------|------|-----|
| All ratings | 1.79 | 1.36 | 174 |
| Rank 1      | 2.03 | 1.51 | 60  |
| Rank 2      | 1.28 | 0.69 | 60  |
| Rank 3      | 2.07 | 1.59 | 54  |

**Listeners agree with R@K.** A mean of 1.79 on a five-point scale says most retrieved clips do not match the caption that retrieved them. That is what Table IX already implied — with R@10 at 0.0339 and a median true-clip rank of 400, the three clips a listener sees usually do not contain the right one. Two independent instruments, one automatic and one human, place this system in the same place, which is stronger evidence than either alone.

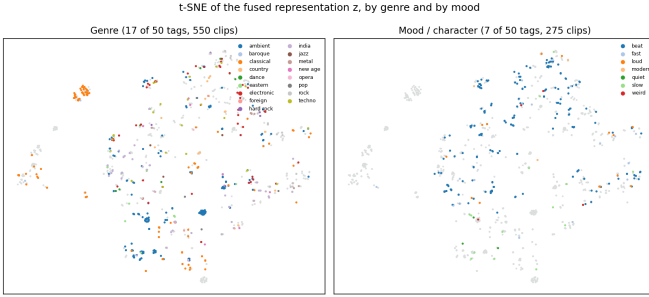
**The model’s own ranking carries no signal for listeners.** Rank 2 scores *below* both rank 1 and rank 3, and the correlation between a clip’s similarity score and its mean human rating is  $r = -0.051$  — indistinguishable from none. So within the top three the ordering is arbitrary to a listener, even though those three were selected out of a 2,773-clip gallery. This is consistent with the picture in Section V-E: the learned

space separates relevant from irrelevant weakly, and does not rank finely within what it has already selected.

**Retrieval is more often plausible than correct.** Fourteen of the 174 ratings were 4 or 5, spread across most queries, while only 0.58% of queries place the true clip at rank 1. A clip can legitimately match a generic caption without being the annotated one, so the human study measures something R@K cannot: the model finds *believable* audio considerably more often than it finds the *intended* audio. For a retrieval system that distinction matters, and it is invisible to recall alone.

**Study design.** Each listener saw one clip per screen with its caption, and the three clips for a caption were presented in a random order, so the ranking under test was never visible. A clip that could not be played was recorded as a gap rather than a low score: one clip was unavailable and all six listeners marked it so, giving 174 ratings rather than 180. Time on task ranged from 6.3 to 17.3 minutes against 5.0 minutes of audio, so no submission was faster than listening allows.

### I. Qualitative analysis of the fused space



**Fig. 5:** t-SNE of the fused representation  $z$  (cross-attention, 384-dim) over 1,500 held-out MagnaTagATune clips, split into the two views the specification asks for. Each clip is coloured by its *primary* tag — the single highest-prevalence tag it carries — not by its full multi-label annotation, and clips whose primary tag falls outside the panel’s vocabulary are drawn in grey. MagnaTagATune’s top-50 list mixes genre, mood, instrument and production descriptors in one flat vocabulary, so the split is ours: the mood panel is a proxy that includes tempo and production words (*fast*, *quiet*, *loud*) alongside affective ones. Broad timbral families separate — classical and vocal-led material occupy distinct regions — but the structure is diffuse rather than cleanly clustered, consistent with the modest Macro-F1 in Table VI.

**Case studies.** Table XIII pairs graph structure with the caption tokens the graph vector attends to, for three held-out clips from distinct tracks.

**TABLE XIII:** Three case studies. “Attn. spread” is the ratio of largest to smallest attention weight over the top-8 tokens; values near 1.0 indicate the collapse discussed in Section 5.3.

| True tags                      | Top predictions          | Attn. spread |
|--------------------------------|--------------------------|--------------|
| drum, electronic, synth, weird | techno, electronic, beat | 1.03         |
| ambient, flute, new age        | drum, india, ambient     | 1.03         |
| beat, electronic, fast, techno | rock, male vocal, vocal  | 1.11         |

The first case is a partial success: an electronic track tagged *drum*, *electronic*, *synth* is predicted *techno*, *electronic*, *beat*, capturing the family if not the exact vocabulary. The third is a clear failure: an electronic/techno clip is predicted *rock*, *male vocal*, *guitar*. In every case the attention spread is within 3% of uniform, so these predictions are driven by the text mean and the graph vector, not by any selective reading of the caption. These case studies come from the shared-learning-rate model; the separate-learning-rate run does attend selectively (peak  $4.19\times$  uniform).

### J. A control: the same fusion on text that describes the audio

Our central Task 3 claim is that the cross-attention collapses because MagnaTagATune’s text is track-level metadata with nothing to attend to, not because the architecture is wrong. That is testable, so we tested it: the identical model, hyperparameters and separate graph learning rate, trained on MusicCaps graphs and captions over the 50 aspects Task 1 uses (3,748/803/804 split).

**TABLE XIV:** The same architecture on two corpora. The F1 rows are *not* a like-for-like comparison — different corpus, labels and difficulty — and are given only so the attention rows, which are comparable, can be read in context.

|                     | MagnaTagATune<br>(metadata) | MusicCaps<br>(descriptions)    |
|---------------------|-----------------------------|--------------------------------|
| Macro-F1 / Micro-F1 | 0.246 / 0.415               | 0.673 / 0.727                  |
| Attention entropy   | 0.695                       | <b>0.393</b>                   |
| Peak over uniform   | $4.19\times$                | <b><math>29.4\times</math></b> |
| Case-study spread   | $1.03\text{--}1.11\times$   | $3.6\text{--}58.5\times$       |

### The collapse is a property of the text, not the architecture.

Given real captions the same cross-attention becomes sharply selective — entropy 0.393 of uniform, peaks near  $30\times$  — where on MagnaTagATune its case-study weights sat within 3% of uniform. That is the control the diagnosis needed.

**The F1 gap is not a win, and we do not report it as one.** MusicCaps aspects appear verbatim in their own captions; Section V-A measures a lexical-match baseline at Micro-F1 0.540 on this exact material, so a model reading them is partly doing substring detection. The attention statistics are the result here, not the accuracy.

The case studies are correspondingly richer than Table XIII’s, which could only align a graph against a track title: a caption describing “digital drums ... two guitars ... an e-bass ... a piano” draws *piano* and *e-bass* into the top four, and each of the three cases recovers two true aspects. One caveat is visible in the weights themselves — the highest-attention tokens are determiners and copulas (*this*, *is*, *the*), not the musical content words. The attention is sharp without being semantic, which is the freeze-BERT lesson of Section V-E arriving from the opposite direction: sharpness is necessary for the mechanism to be doing anything, and is not evidence that it is doing the right thing.

**Graph paths.** Each case study records a walk through its segment graph, so a route can be read against the caption:

it starts at the highest-degree segment and steps to unvisited neighbours. Segment  $k$  covers seconds  $1.5k-1.5(k+1)$ , so the walk  $[1, 0]$  names 1.5–3.0s then 0–1.5s. It is an illustrative walk and nothing more: it is unweighted, because the edge weights the cache stores were not carried into the graphs these case studies came from, and its stopping after two nodes says only that the greedy walk ran out of unvisited neighbours — not that the graph lacks longer paths, which would need a path statistic we did not compute.

**Retrieval behaviour.** Across the official-split gallery of 2,773 clips the median true-clip rank is 527, placing the correct clip in the top 19% of the gallery on average — clearly better than chance ( $8.0\times$  random at R@1), but far from usable retrieval. The model appears to learn coarse acoustic-descriptive correspondence (recording quality, instrumentation family, tempo band) without the fine discrimination that rank-1 retrieval demands.

## VI. ENGINEERING FINDINGS

Several results in this report depended on catching pipeline defects that produce plausible-looking numbers. We record them because each was silent.

**Near-complete graphs.** Our first graph build produced 19.4-node graphs with average degree 18.15 — effectively complete. A GNN over a complete graph computes the same neighbourhood for every node, so message passing reduces to global mean pooling and the architecture becomes irrelevant while still training and reporting sensible-looking accuracy. The cause was that raw MFCC descriptors share a large positive mean, giving median pairwise cosine similarity 0.9971. Only after z-scoring within each track did the threshold behave as intended.

**The opposite failure.** Having fixed the scaling, the threshold constant was left at its old value, producing average degree 1.90 — a bare temporal chain with *zero* similarity edges. Both failures produce trainable models; neither announces itself.

**A silent truncated cache.** Serialising the preprocessed MagnaTagATune cache wrote a zero-byte file without raising, because the log-mel component alone was 3.49 GB of a 3.6 GB structure. Since only the CNN baseline consumes log-mel, it is now optional.

**Threshold selection.** Multi-label probabilities under prevalence-weighted BCE do not centre at 0.5; our tuned thresholds ranged from 0.7 to 0.8. Evaluating at 0.5 would have understated every Task 1 and Task 3 result substantially.

**A segfault that looked like a deadlock.** The graph builder appeared to hang: it froze at 4,000/8,000 FMA tracks and, on another run, at 10,500/21,318 MagnaTagATune clips, with every worker asleep on a futex at  $\approx 0$ s of CPU. The real event was a *segmentation fault* inside a numba guvectorize kernel reached through librosa [10] `chroma_stft`  $\rightarrow$  `estimate_tuning`  $\rightarrow$  `piptrack`, caused by a numba 0.67 / numpy 2.1 ABI mismatch in the GPU image. The deadlock appearance is a property of `multiprocessing.Pool`: it silently

respawns a worker that dies mid-task and simply never delivers that task’s result, so the parent blocks forever. Pinning the numba/llvmlite pairing fixed the crash, and replacing the pool with `ProcessPoolExecutor` — which raises `BrokenProcessPool` immediately on worker death — converted an hour of silent stalling into an instant diagnosable failure. Throughput also rose from 8.8 to 16.2 tracks/s.

**Two split defects.** Our Task 1 and Task 4 MusicCaps experiments originally used random splits even though the metadata ships an `is_audioset_eval` flag defining the official partition; the corrected official-split results are materially worse and are what this report now carries. Separately, the Task 1 random baseline estimated per-tag prevalence from *test* labels, which leaks test statistics into a baseline; it now uses training labels only.

**Checkpoint policy.** Task 1 and Task 3 evaluated whichever epoch happened to be last rather than restoring the best validation checkpoint. This was not cosmetic: Task 3’s GNN-only ablation peaks at epoch 1 and decays to near-zero validation Macro-F1 by epoch 8, and correcting it changed the ranking of the Task 3 ablation table.

We mention these not as incidental debugging notes but because most of them would have yielded a complete, publishable-looking results table built on a model, a split, or a baseline that was not doing what its description claimed.

## VII. DISCUSSION AND LIMITATIONS

Three limitations bound what these results can claim.

**The text is impoverished.** For MagnaTagATune our BERT branch sees only “title | album”. This is the source of both the 0.673 ceiling and, plausibly, the attention collapse: there is little in a track title for a graph vector to select between. A corpus pairing audio with genuine descriptive text at clip granularity — as MusicCaps does — is the right setting for fusion, and our Task 1 results support this, with stripped-caption Micro-F1 of 0.488 versus 0.275.

**The graph branch was under-trained, and fixing that was decisive.** GNN-only reaches 0.123 Macro-F1 with 31.8K parameters against BERT’s 66.4M, sharing one learning rate. We ran both diagnostics this predicted. A separate graph-branch learning rate of  $10^{-3}$  raised cross-attention Macro-F1 from 0.192 to 0.246 and Micro-F1 from 0.325 to 0.415 while cutting attention entropy from 1.000 to 0.695 of uniform; freezing BERT instead produced the sharpest attention of any run but the worst Macro-F1. The conclusion is that the shared learning rate, not the architecture, was the binding constraint — and that attention sharpness alone is not a sufficient indicator of a working fusion. A five-seed paired sweep confirms the effect ( $+0.067 \pm 0.023$  Macro-F1,  $p = 0.003$ , positive on every seed). The natural next step is to tune the graph-branch learning rate properly rather than accepting the first value we tried.

**Segment graphs may be the wrong structure, and the alternative is worse.** Our nodes are fixed 1.5s windows. Table III shows the chord-transition graph the specification also names performs 0.110 Macro-F1 below it, so the segment

graph is the better of the two structures we built — but a 19-node graph over uniform windows may still encode little beyond a smoothed timeline, and beat-synchronous segmentation, which we implement but did not train on, remains untested. The chord result suggests the limitation is in the node features rather than in message passing.

**The human study is small, and its scale is its limit.** Six listeners and 174 ratings is enough to establish that retrieved clips mostly do not match their captions, and enough to show that the model’s ranking does not predict human judgement. It is not enough to estimate either quantity precisely, to model listener effects, or to detect a small difference between ranks. Listeners were not screened for musical training or hearing, and with six raters no such split would have been interpretable. We therefore report the aggregate and the per-rank means, and make no claim resting on a difference smaller than the spread between listeners, whose means range from 1.28 to 2.48.

**Single seeds.** Every result in this report except the Task 3 learning-rate comparison and the Task 1 curves comes from one seed. Our own sweep puts seed-to-seed spread at  $\pm 0.023$  Macro-F1, which exceeds several of the margins in Table VI; those rankings should be read as indicative. The data-scale points in Table X are likewise single seed, which is why we treat the flattening in the last interval as noise rather than saturation.

## VIII. CONCLUSION

We implemented all four tasks. Our first conclusion about fusion was negative and turned out to be an optimisation artefact: under a single learning rate the cross-attention collapses and fusion loses, and once the non-BERT parameters train at their own rate the same architecture beats its text branch. We regard the diagnosis, and the correction of our own claim, as the contribution: a measured performance ceiling from duplicated inputs, a measured artist leak in a standard split, a measured attention collapse, a multi-task term that proved roughly free rather than costly once our own checkpoint policy was corrected, and a human study that agrees with the automatic metric while showing the ranking beneath it to be arbitrary. Each of these would have been invisible had we reported only headline F1 against no baseline — and in one case (raw-caption tagging) doing so would have overstated our result roughly fivefold. Cross-modal fusion for music context understanding remains plausible; this work delimits the conditions under which it can be demonstrated.

## REPRODUCIBILITY

Code, seeded configurations, per-run metrics, 155 example graphs (50 FMA, 105 MagnaTagATune, each as both `.pt` and `.json` with an index manifest), the t-SNE projection and all training logs are in the project repository. Headline tables and claims are generated from `results/metrics.json` into `report/generated_metrics.tex` by `src/make_report_tex.py`; the report includes figures directly from `results/plots/`. Missing required values stop generation, preventing stale hand-copied tables or plots

from compiling unnoticed. The train/validation/test manifests for all three corpora are committed under `data/splits/`, so the exact partition behind every number can be inspected without re-running preprocessing. Training requires a GPU; every evaluation, plot and notebook in the repository runs on CPU in seconds from the committed artifacts. The human study ran as a web application whose source, stimuli and per-listener submissions are all committed, so it can be repeated with new listeners rather than merely described. The Task 1 MusicCaps checkpoint is downloadable with a published SHA-256; the download, checksum check and a single inference call were verified from a clean copy, and `artifacts/checkpoints/README.md` carries the commands.

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